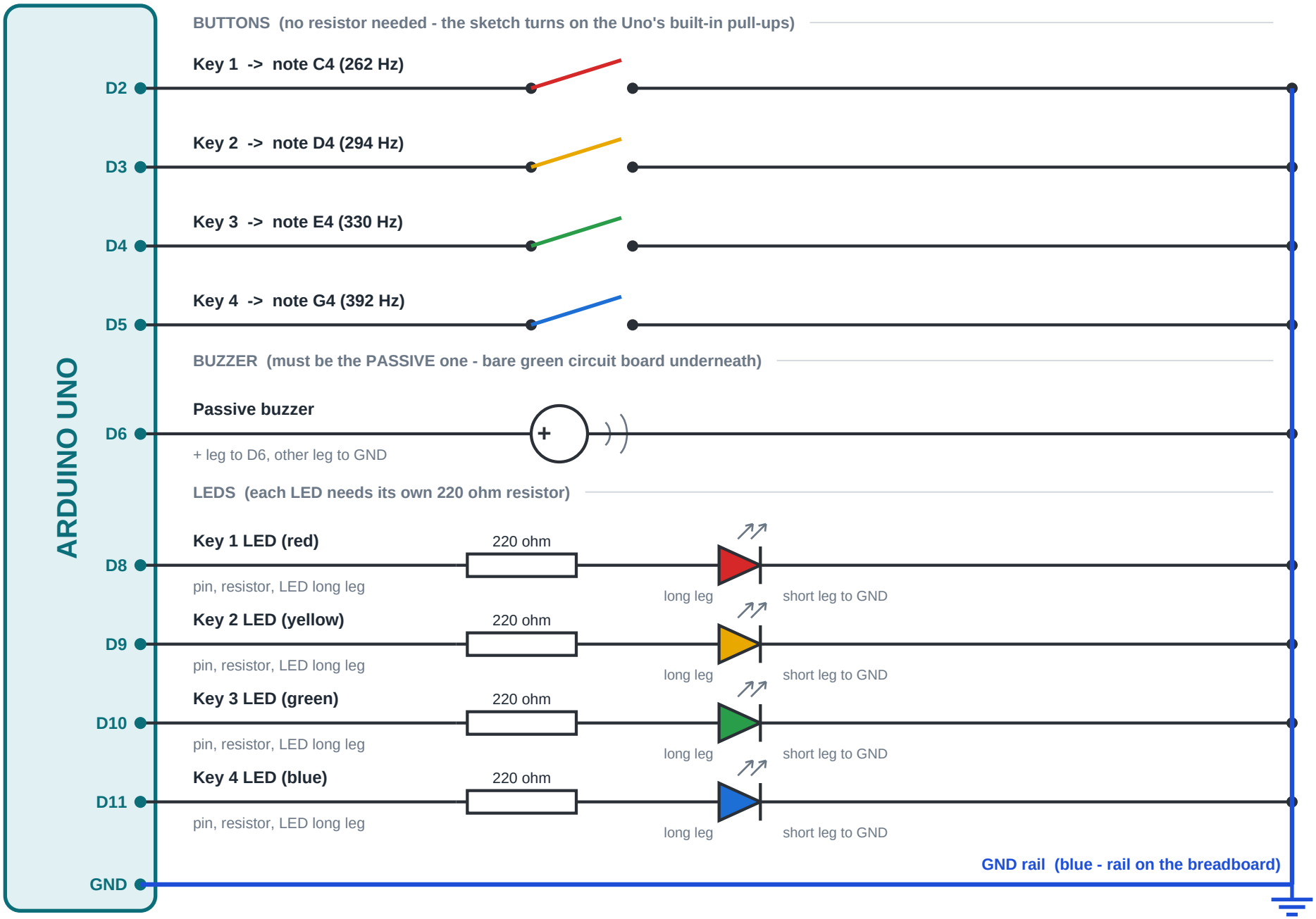
Mini Piano - Wiring Schematic

Arduino UNO | 4 buttons, 4 LEDs, 1 passive buzzer | Sketch: src/main.cpp



Every GND connection goes to the same place: one jumper from the Uno GND pin to the breadboard - rail.

Mini Piano - Parts & Connections

Everything on page 1 as lists, plus a few tips.

Parts

Qty	Part	Notes
1	Arduino UNO + USB cable	Powers everything. No battery needed.
1	Breadboard	Run a jumper from Uno GND to the blue (-) rail.
4	Push buttons	Straddle the centre gap of the breadboard.
4	LEDs	Red, yellow, green, blue. Long leg is + (anode).
4	220 ohm resistors	Colour bands: red, red, brown.
1	Passive buzzer	Bare green circuit board underneath. Not the one with a sticker.
~14	Jumper wires	One per button, LED and buzzer pin, plus GND.

Connections

Uno pin	Goes to	Then to
D2	Button 1	GND
D3	Button 2	GND
D4	Button 3	GND
D5	Button 4	GND
D6	Buzzer (+) leg	GND
D8	220 ohm, LED 1 long leg	LED short leg, then GND
D9	220 ohm, LED 2 long leg	LED short leg, then GND
D10	220 ohm, LED 3 long leg	LED short leg, then GND
D11	220 ohm, LED 4 long leg	LED short leg, then GND

The notes

Button	Note	Pitch
1	C4	262 Hz
2	D4	294 Hz
3	E4	330 Hz
4	G4	392 Hz

If something is wrong

LED stays dark: flip it around. The long leg goes toward the resistor.

Button always seems pressed: a push button has 4 legs and the two on the same side are joined inside. Turn it 90 degrees so it spans the centre gap.

No sound, or just clicks: you have the active buzzer. Swap in the passive one.

Nothing at all: check the Uno GND jumper reaches the blue rail.

Two keys at once: only the lowest-numbered key plays. That is by design.